Introduction to Machine Learning

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Content

- Motivation
- Definitions and problems
- Applications

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- **Motivation**
- Definitions and problems
- Applications

Motivation

- Why machine learning is important?
  - Growing volumes and varieties of available data, computational processing that is cheaper and more powerful, and affordable data storage.
  - Physical, biological, financial phenomenon, etc.: too complex to be described in a deterministic way
  - Machine learning techniques
  - Computers can learn without being programmed to perform specific tasks
  - Voluntary to replace human experts

Motivation

- Why machine learning is important?
  - Adapt to the environment
  - Deal with new problems
  - Data analysis
  - Pattern recognition
  - Computer vision
  - Control robots, self-driving cars
  - Text mining, chatbots
  - Information retrieval
  - Game, etc

Content

- Motivation
- <u>**Definitions and problems**</u>
- Applications

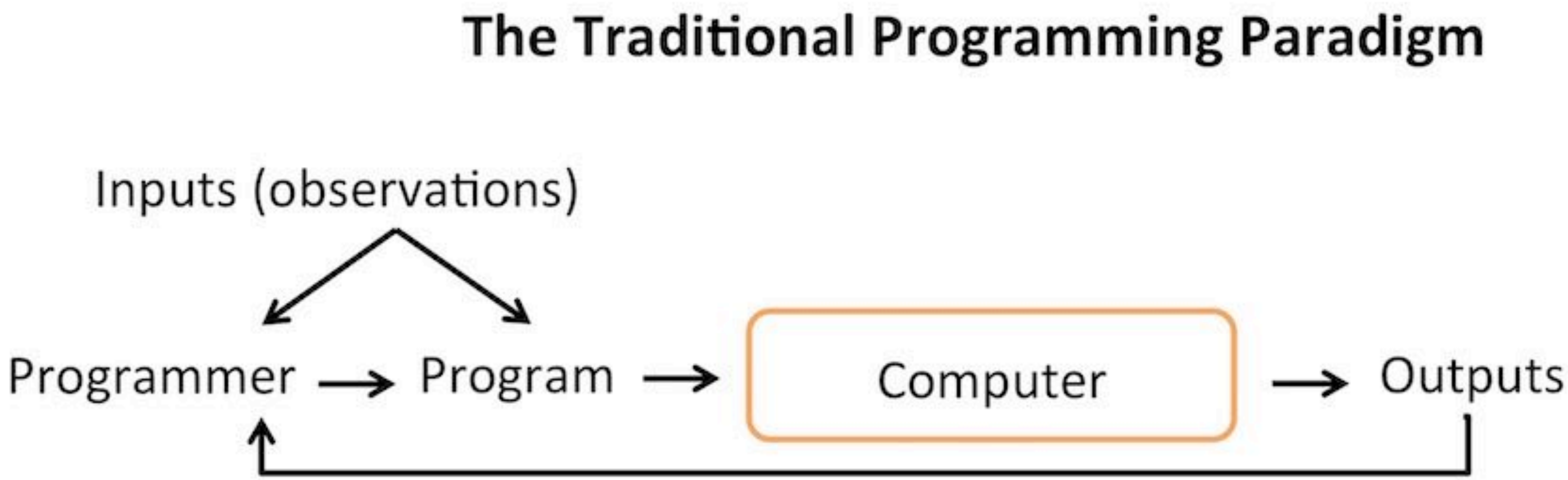
Definitions

- Wikipedia
  - **Learning** is the process of acquiring new, or modifying existing, knowledge, behaviors, skills, values, or preferences

- Arthur L. Samuel, AI pioneer, 1959
  - **Machine learning** is the field of study that gives computers the ability to learn without being explicitly programmed

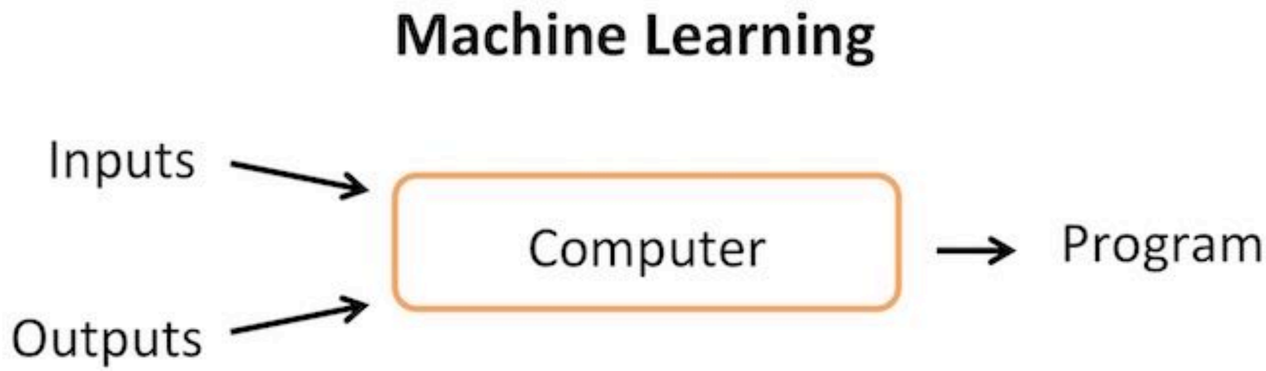
- Tom Mitchell, Professor at Carnegie Mellon University
  - A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P, if its performance at tasks in T, as measured by P, improves with experience E

Definitions



Definitions

Machine Learning is the field of study that gives computers the ability to learn without being explicitly programmed
– Arthur Samuel (1959)

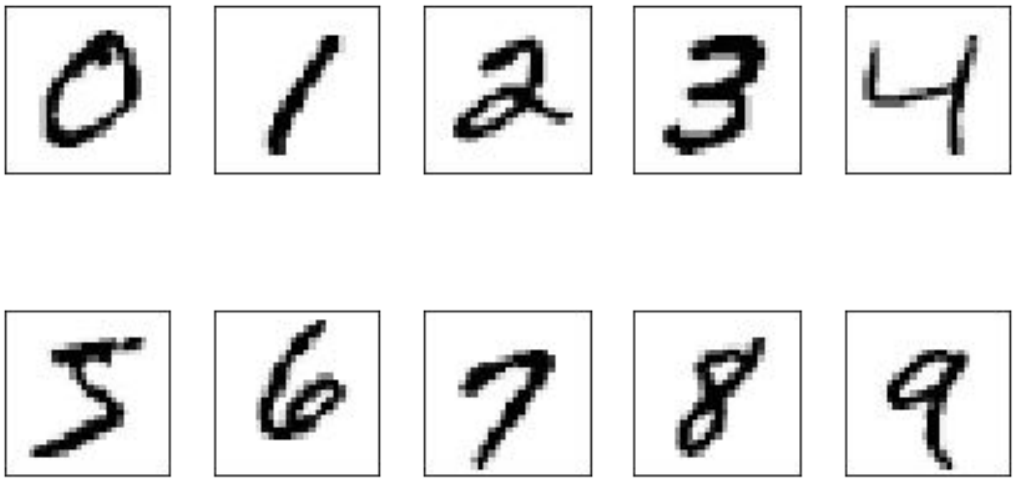


Definitions

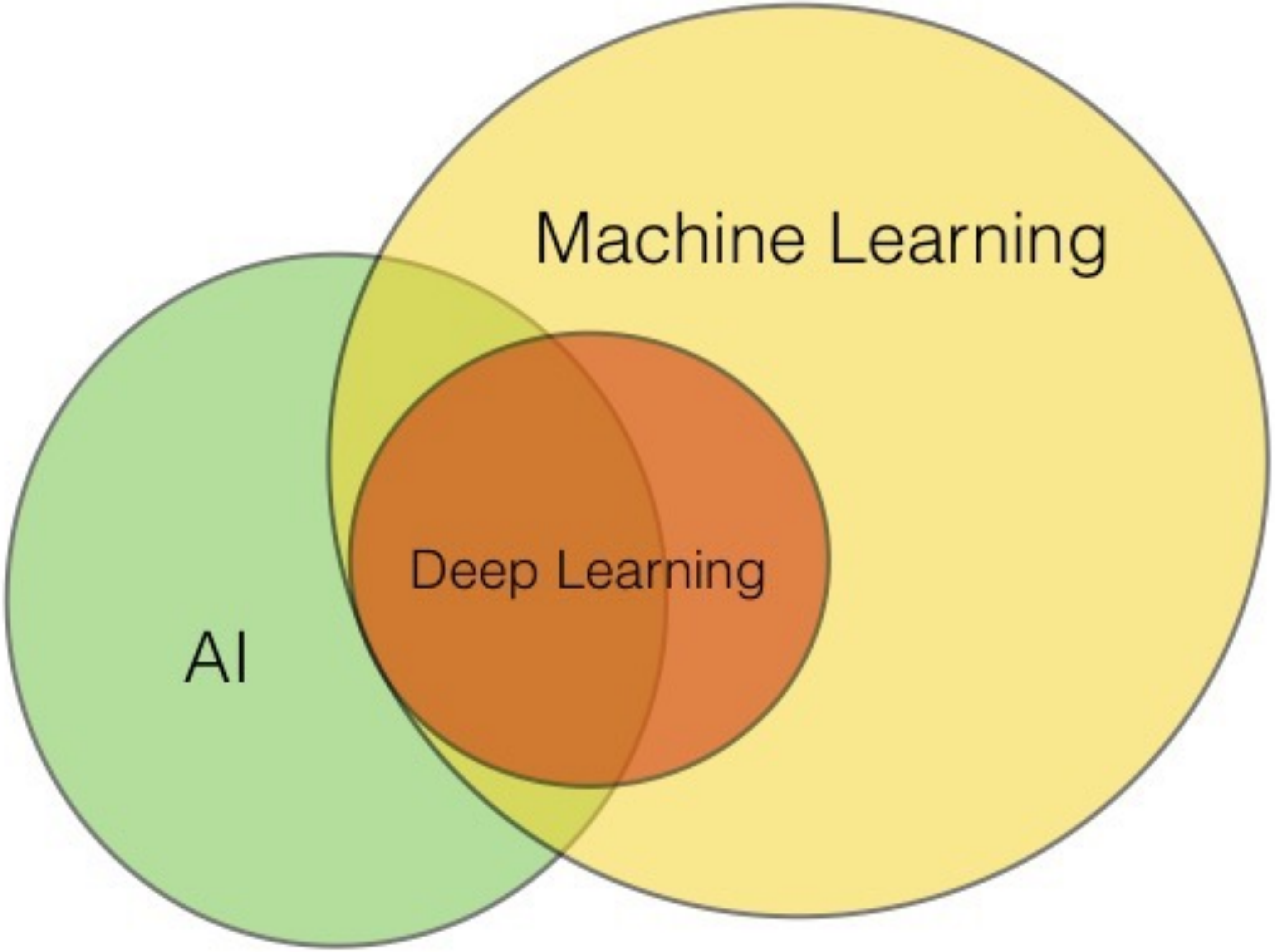
"A computer program is said to **learn** from experience E with respect to some class of tasks T and performance measure P, if its performance at tasks in T, as measured by P, improves with experience E."

— Tom Mitchell, Professor at Carnegie Mellon University

Handwriting Recognition Example:



Machine Learning, AI, Deep Learning



Problems

- Supervised learning
  - Given a set of m training examples

    $D_{Train} = \{(X_1, Y_1), \ldots, (X_m, Y_m)\}$ such that

    $X_i \in R^n$ is the feature vector of the i-th example and

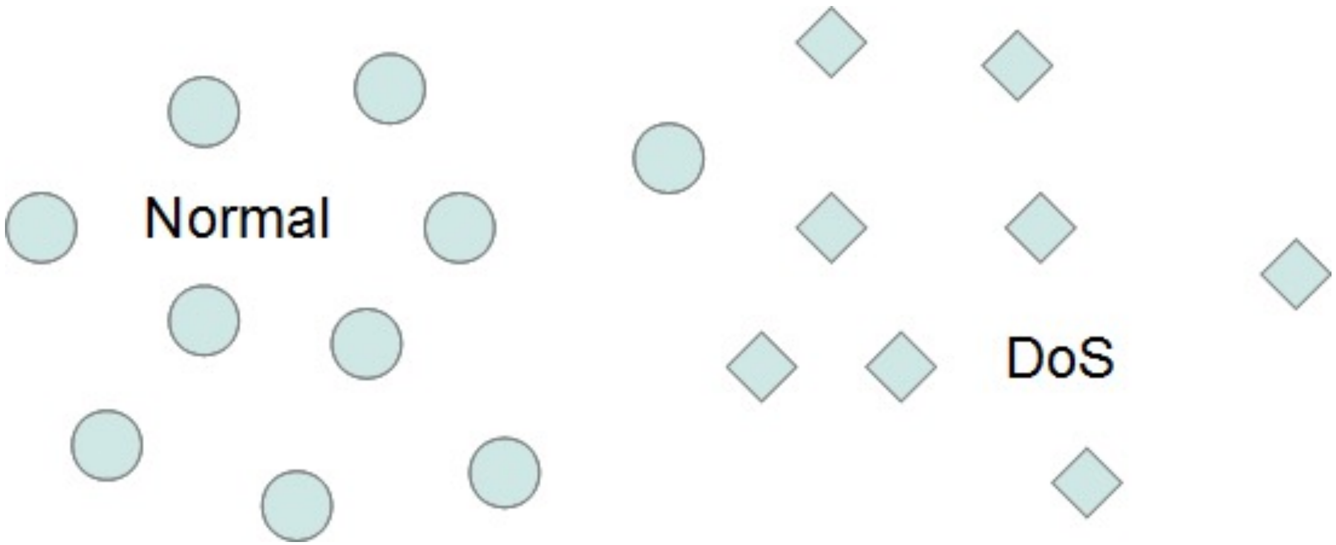
    Y_i is its label (i.e., target)

  - A learning algorithm tries to find a function $\phi: X \rightarrow Y$

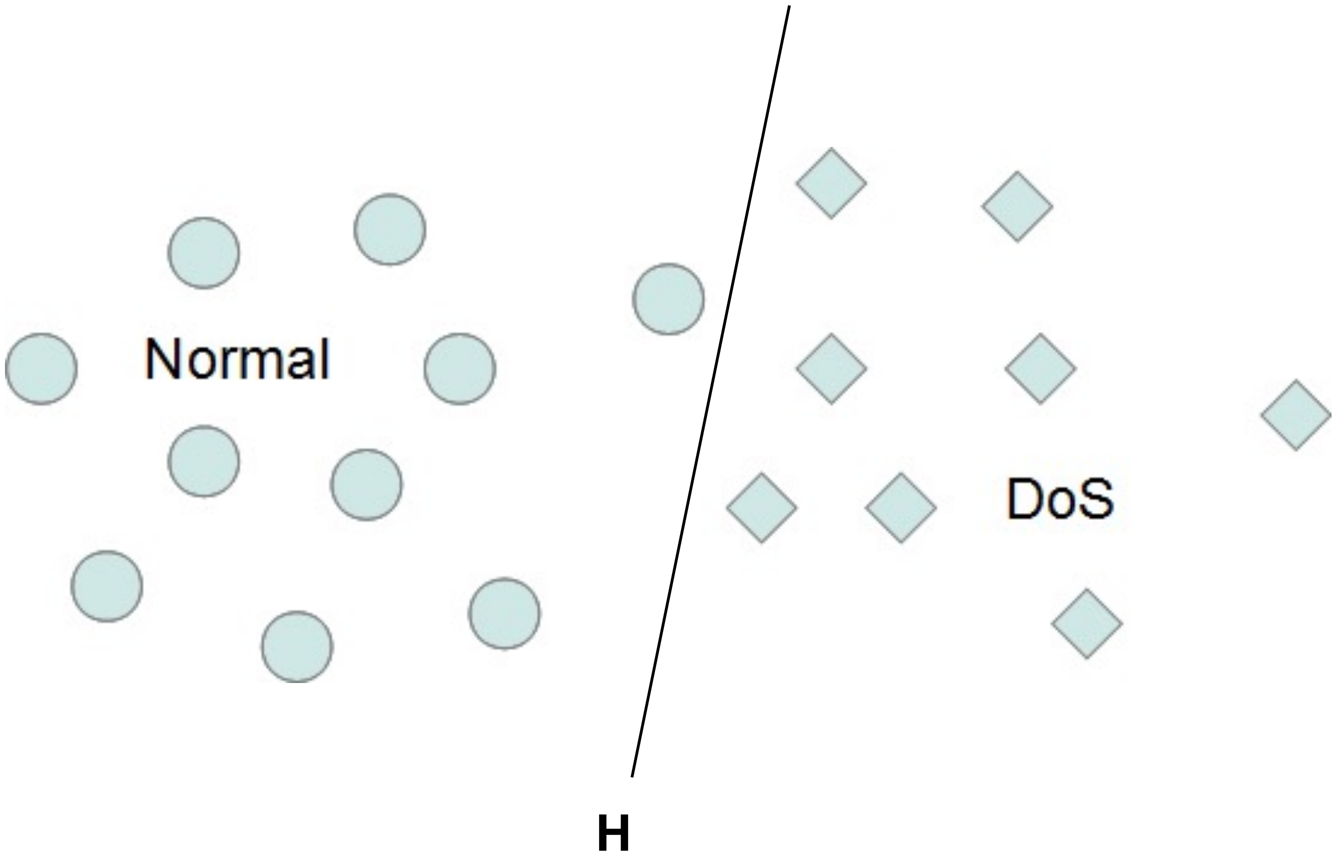
    $Y = \phi(X) + \varepsilon$

  - Methods: decision trees, neural networks, support vector machines, k nearest neighbors, naive Bayes, etc

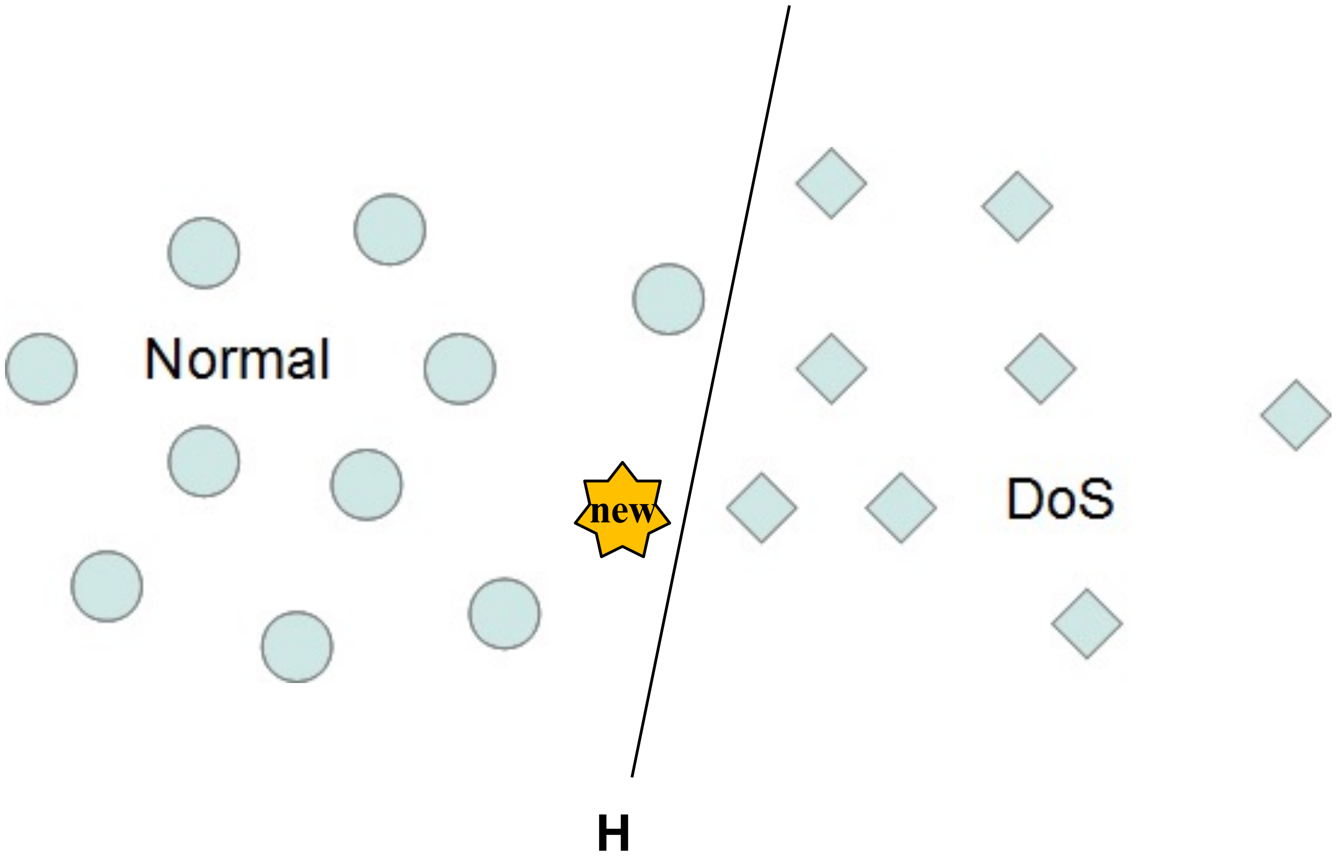
Classification problem



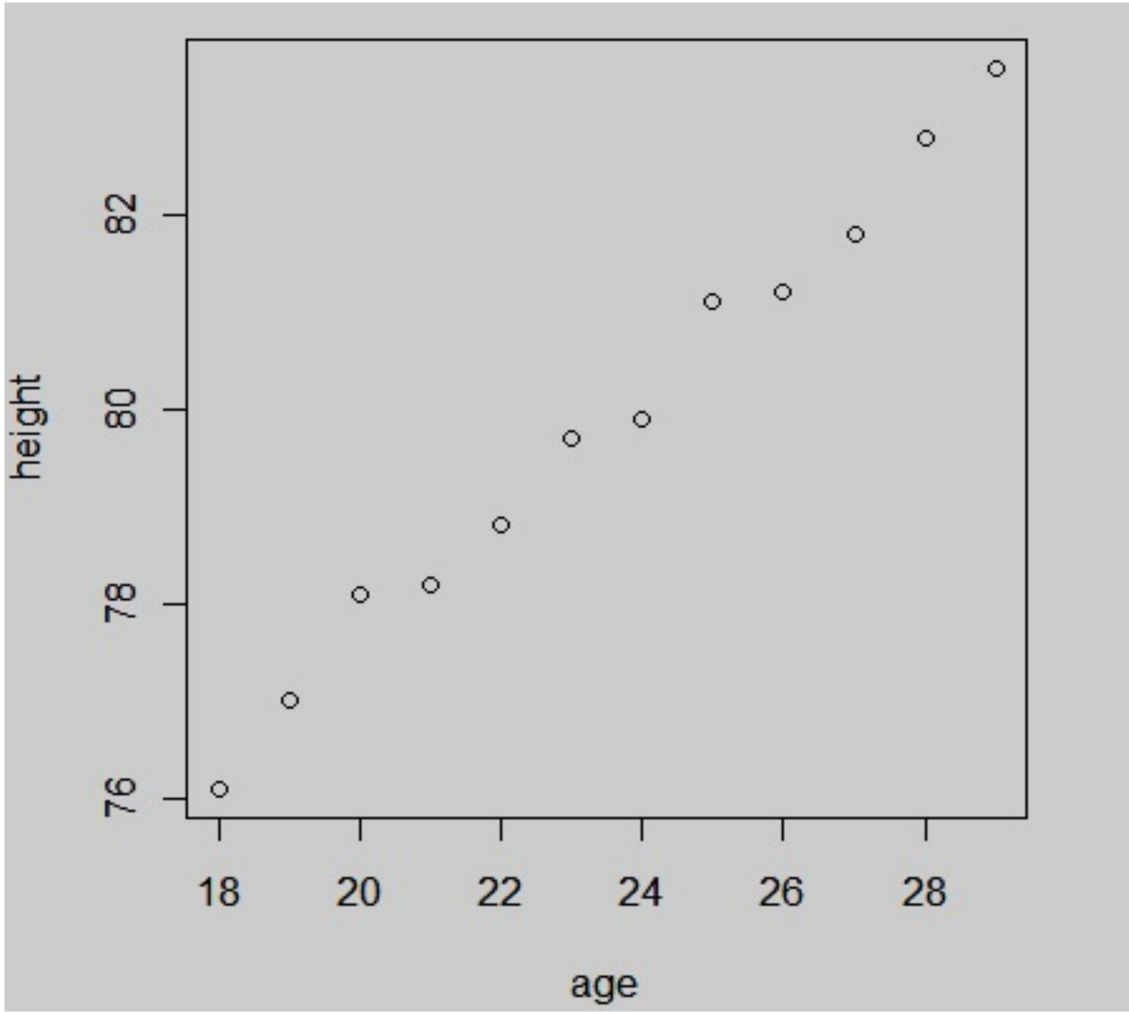
Classification problem



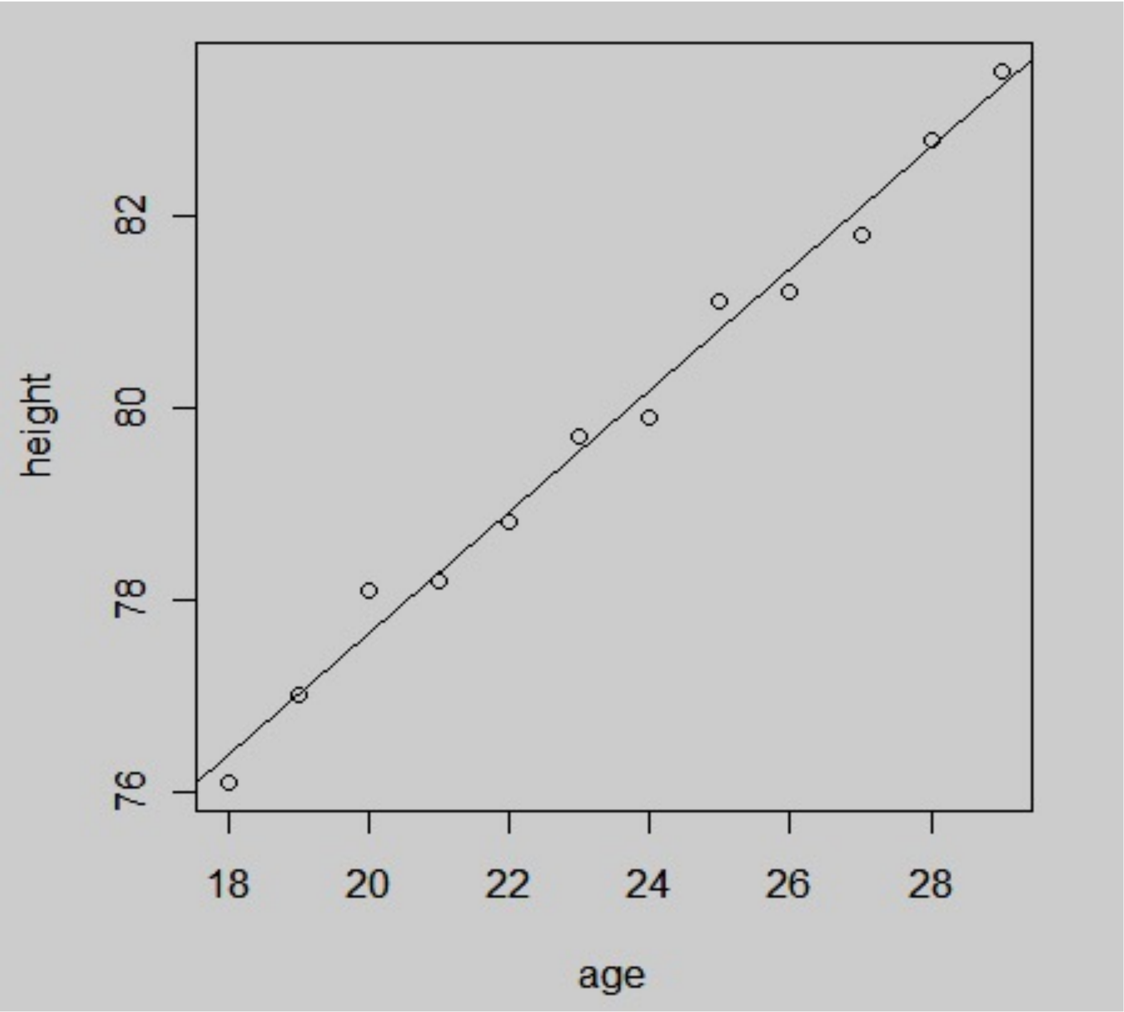
Classification problem



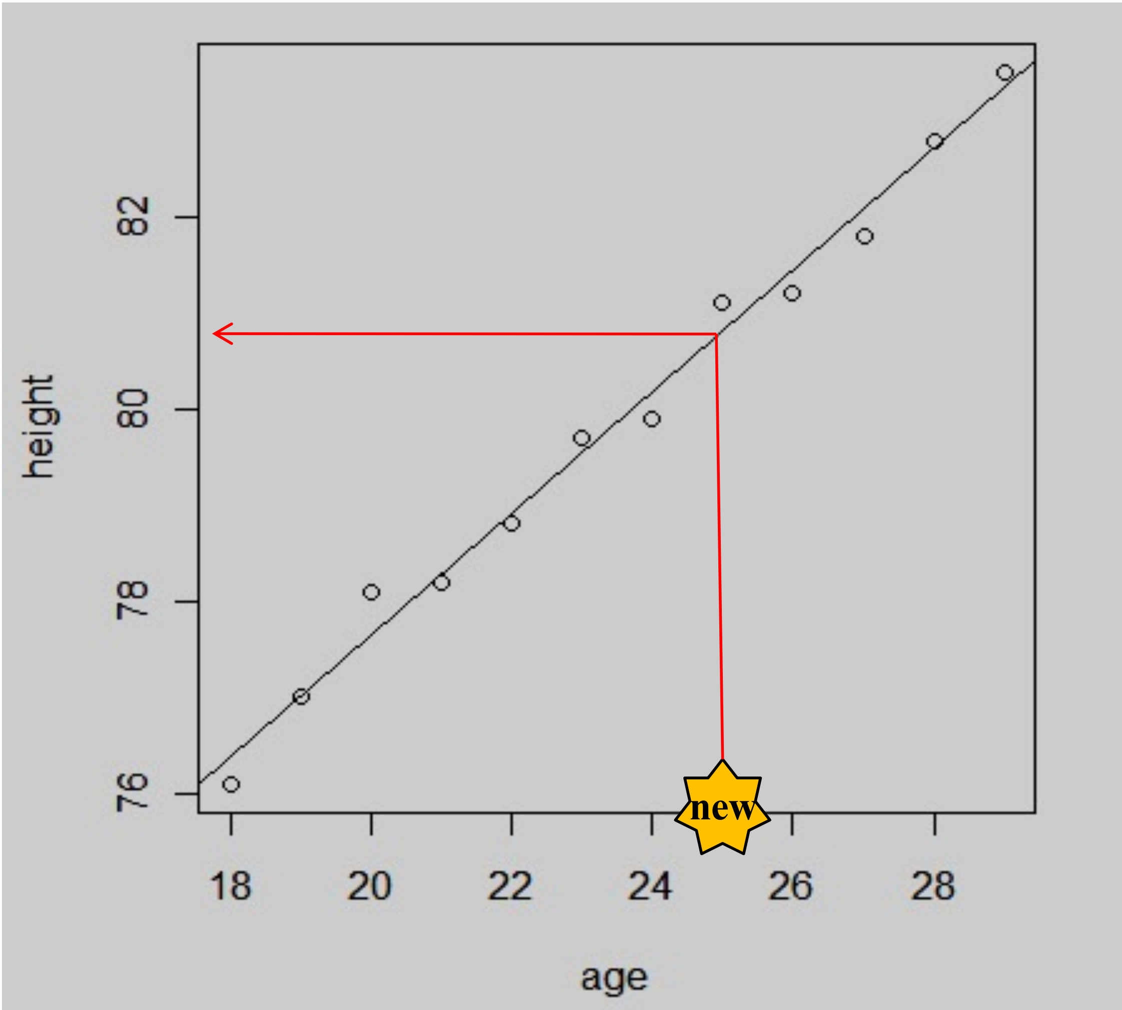
Regression problem



Regression problem



Regression problem



Problems

- Unsupervised learning
  - Given a set of *m* training examples without labels

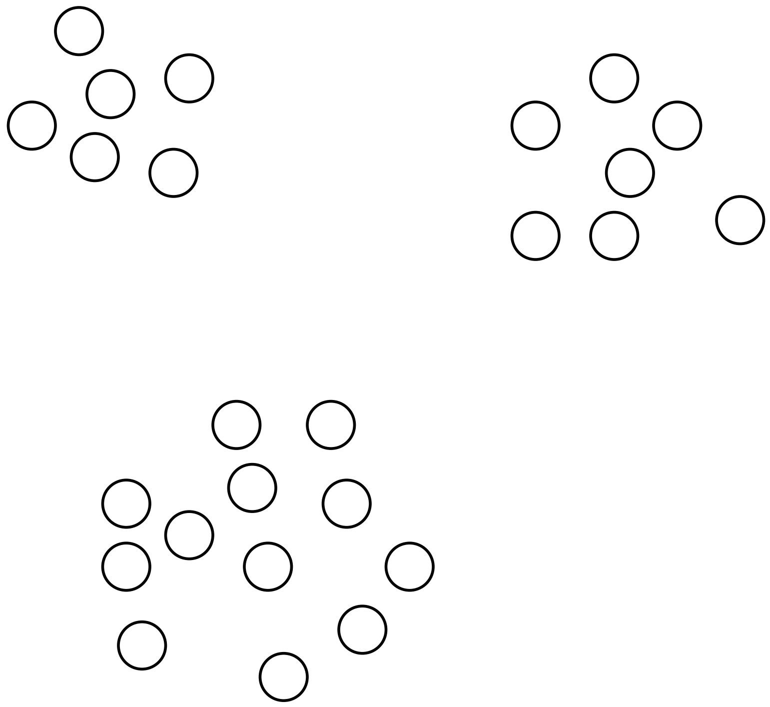
    $D_{Train} = \{X_1, \ldots, X_m\}$ such that

    $X_i \in R^n$ is the feature vector of the *i*-th example

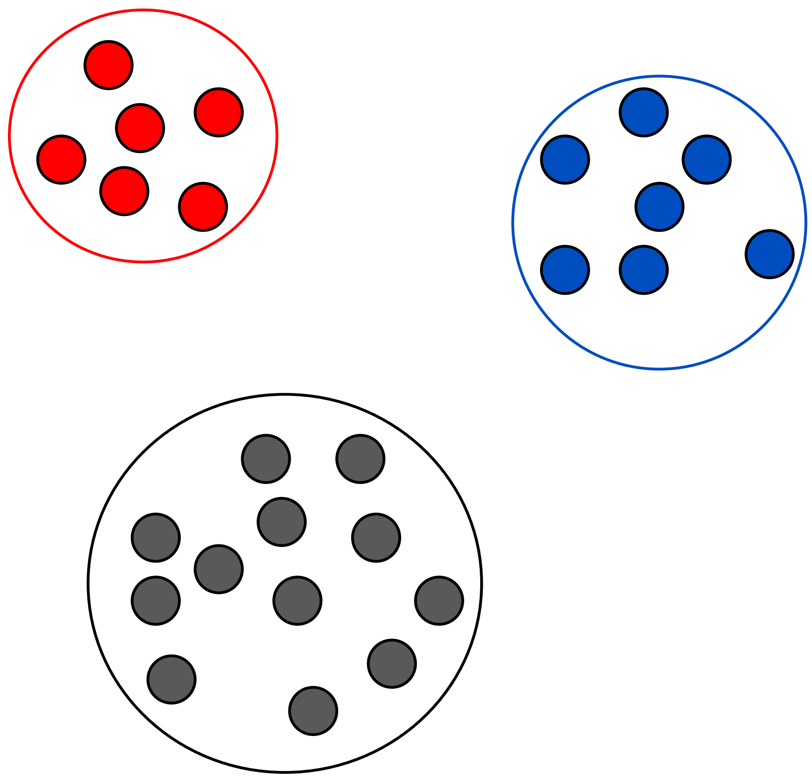
  - A learning algorithm tries to find the hidden structure (i.e. clusters) in unlabeled data according to similarities, patterns and differences without any prior training of data

  - Methods: hierarchical clustering, *k*-means, etc

Unsupervised learning: clustering



Unsupervised learning: clustering



Challenges

- Statistical techniques
  - *m, n* are reasonable; under assumption of the linear model and the data distribution is known (Gaussian, Binomial, Poisson)
- It seems to be very difficult
  - Small samples
  - Unknown data distribution
  - Unknown target function
  - Non-linear models
  - Very large data

Challenges

- Method selection
  - Many learning algorithms
  - No free lunch in machine learning
  - Given a problem => model selection
  - Empirical test, tuning hyper-parameters, comparison
  - Training time, testing time, model quality

Content

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- Definitions and problems
- **Applications**

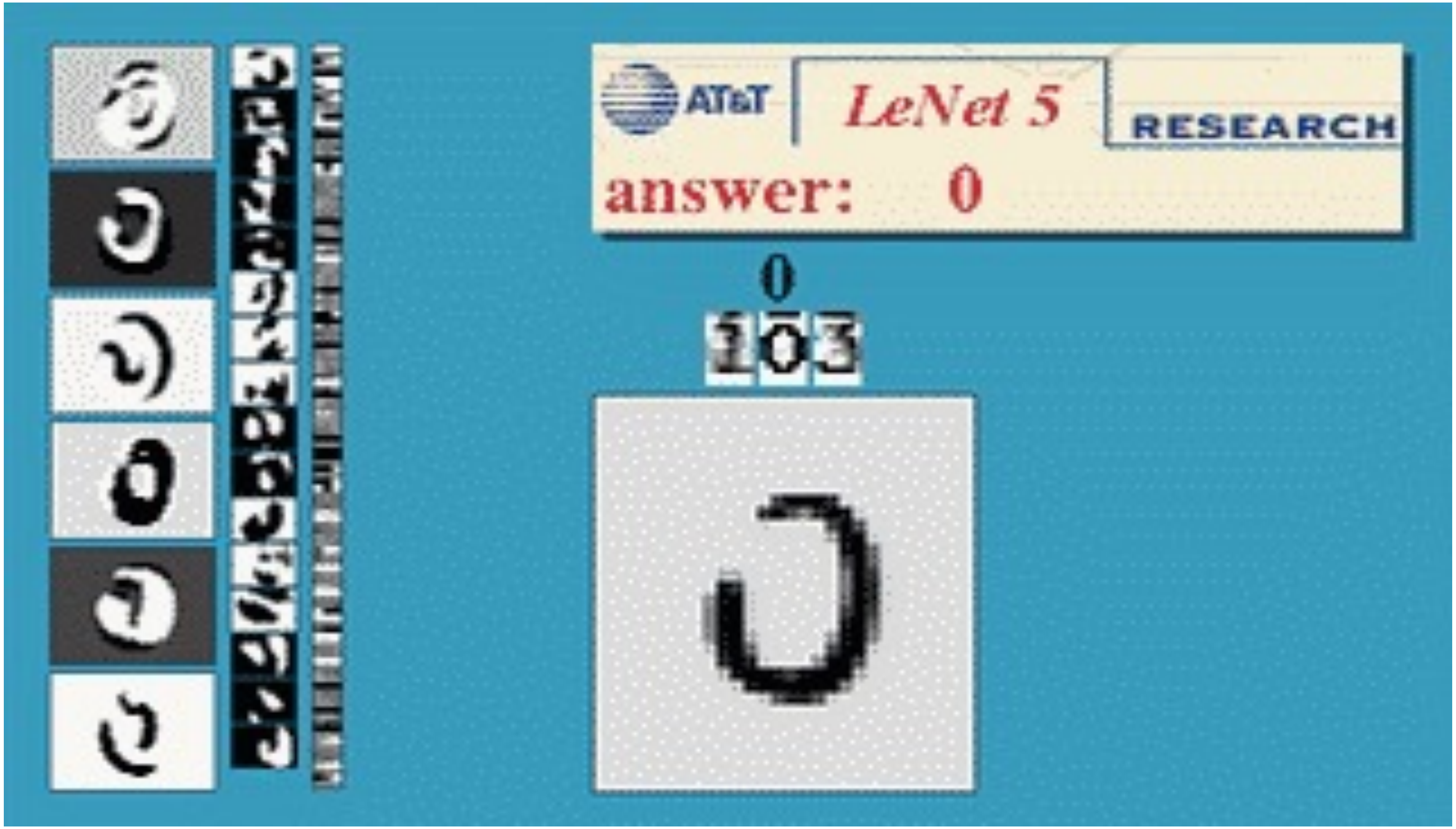
Applications

- Bioinformatics
- Gene expression classification
- Diagnostic support system
- Computer vision
- Handwriting character recognition
- Speech recognition
- Fraud detection
- Financial market analysis

Applications

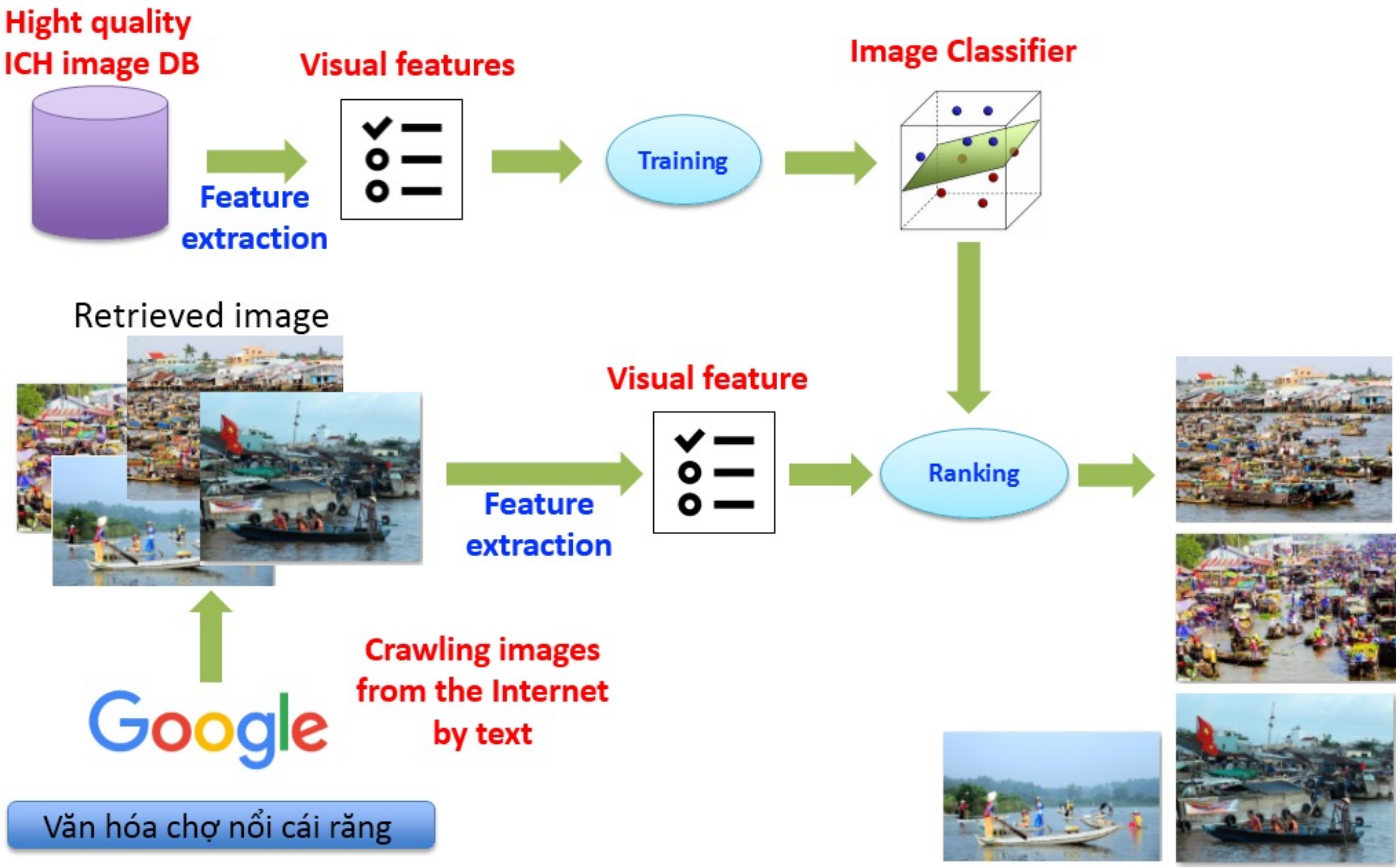
- Information retrieval
- Analyzing and indexing images and vidéos
- Content-based image retrieval
- Games
- Natural language processing
- Recommendation system
- Sentiments analysis
- Robotics
- Self-driving cars
- Cyber-security
- etc.

Applications

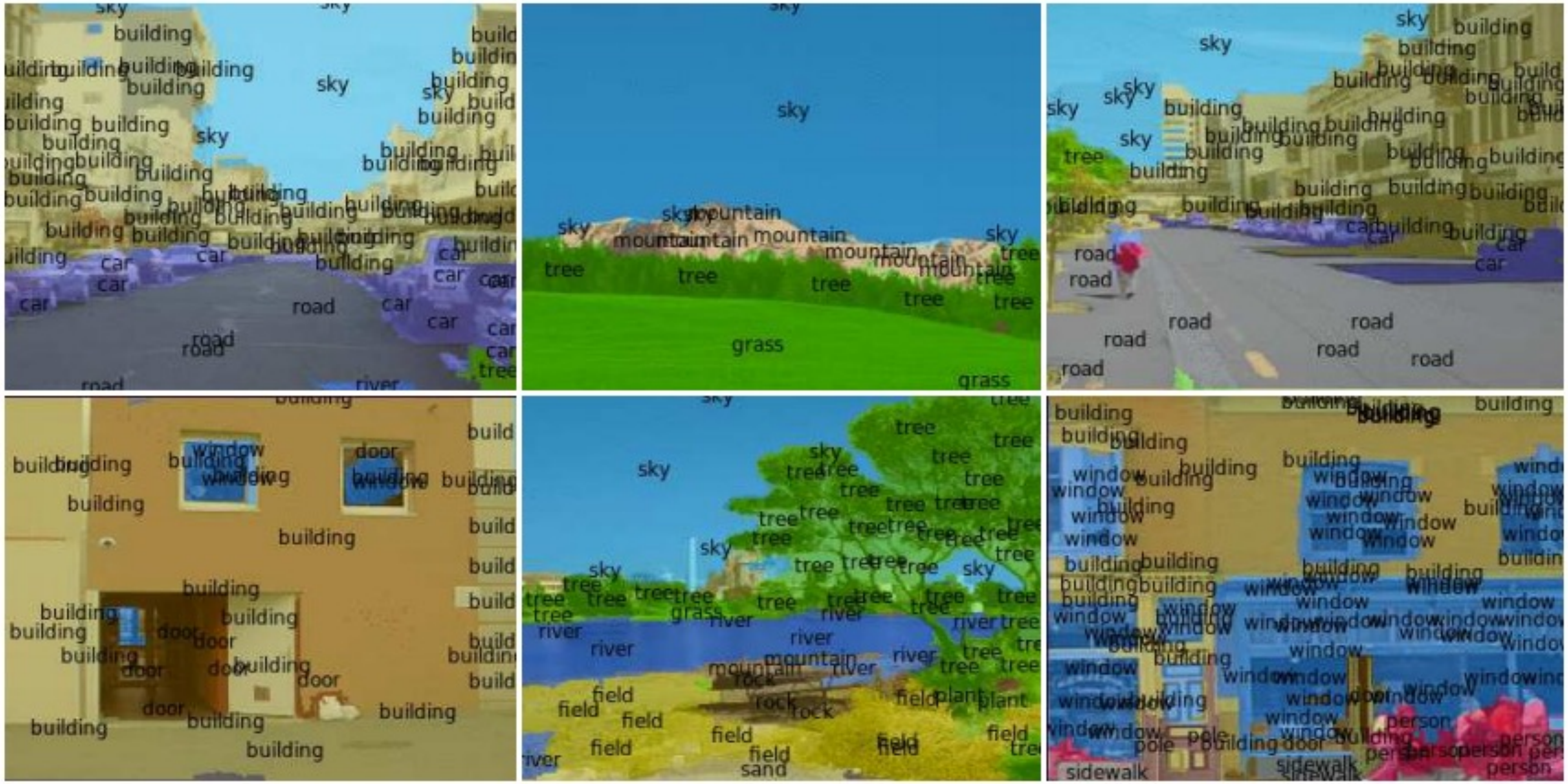


Applications

Applications

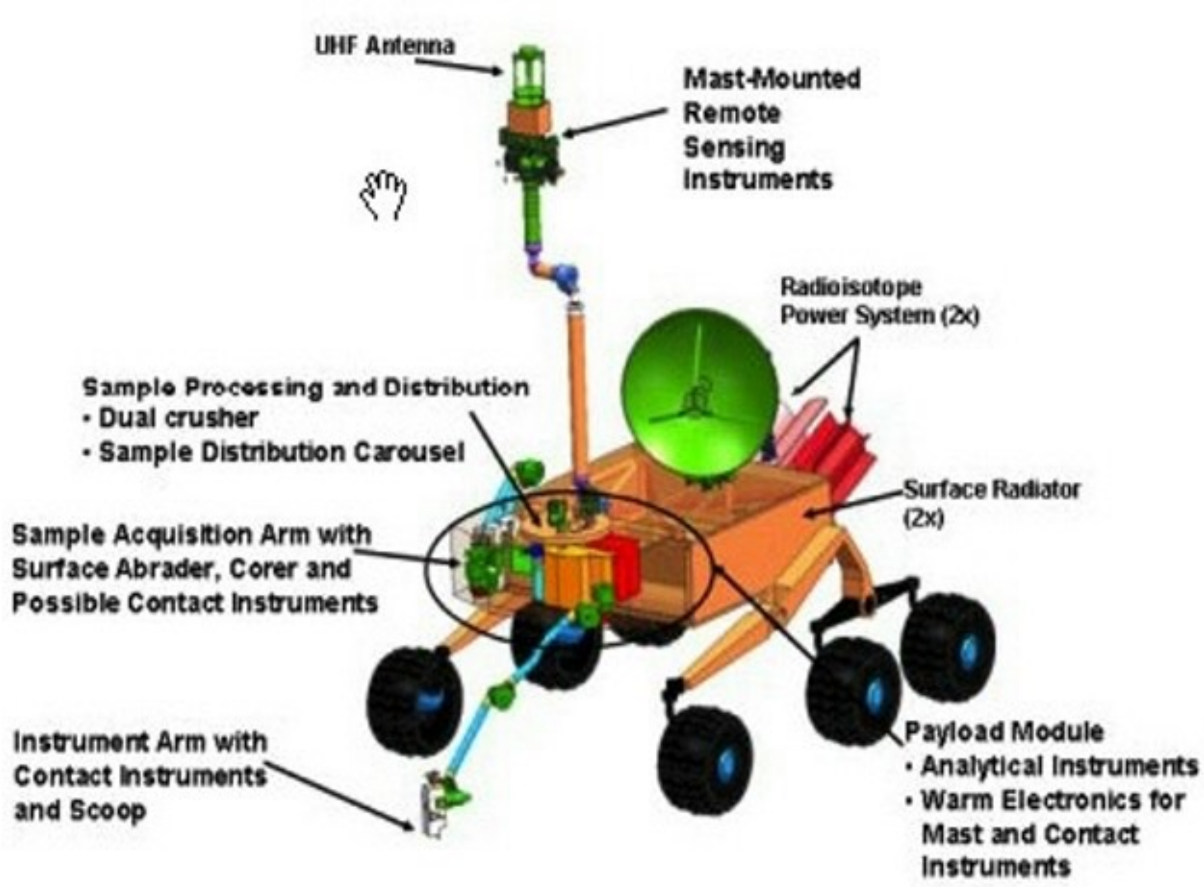


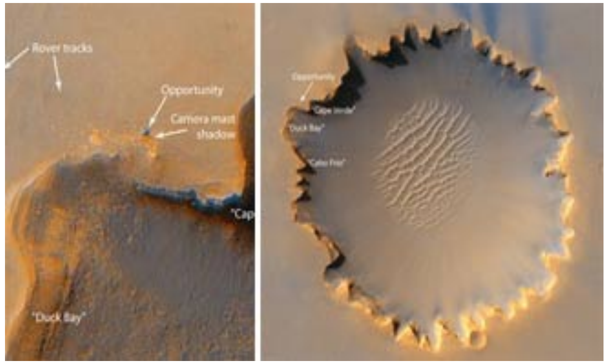
Applications



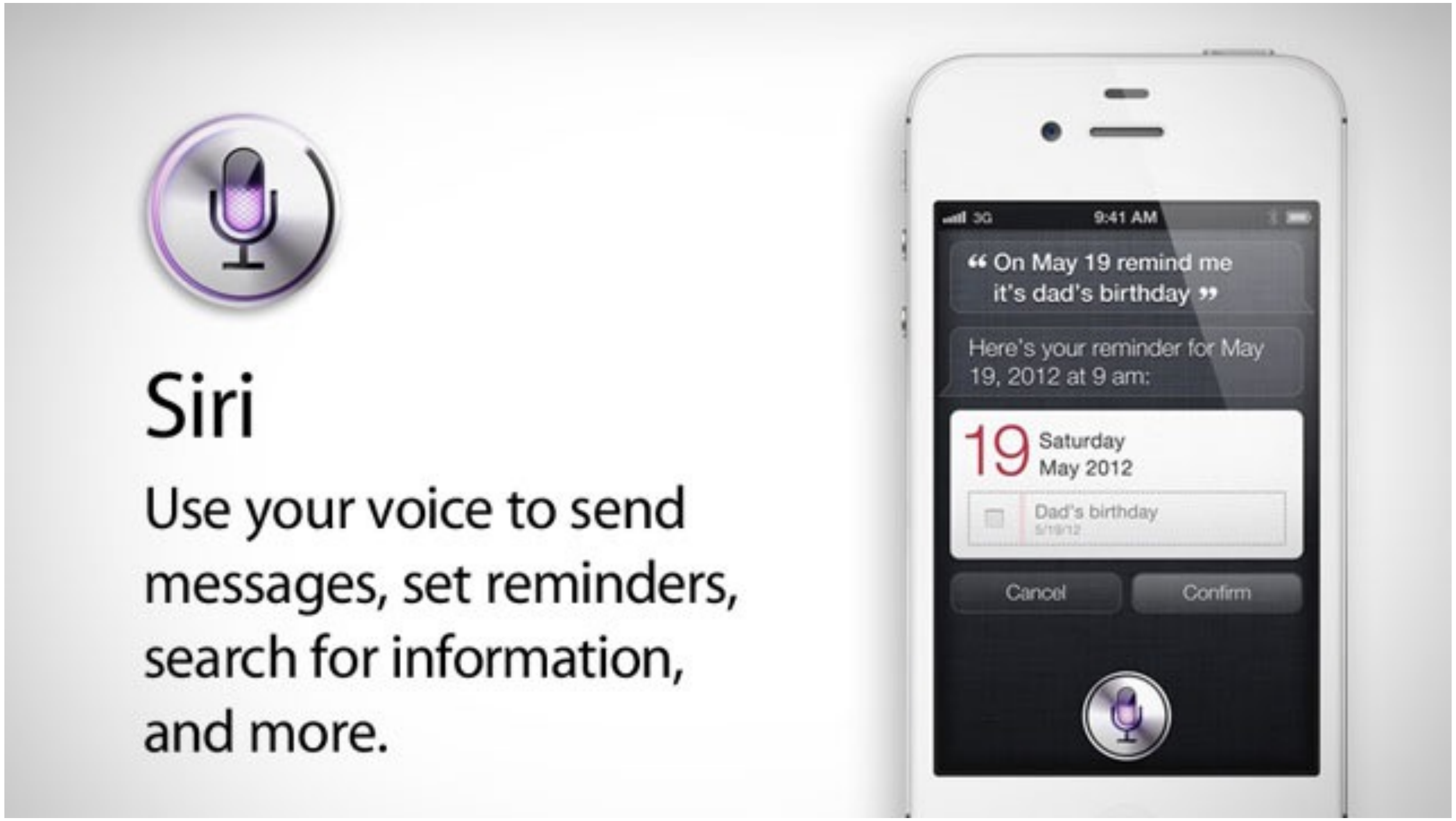
Applications

Applications





Applications



Applications

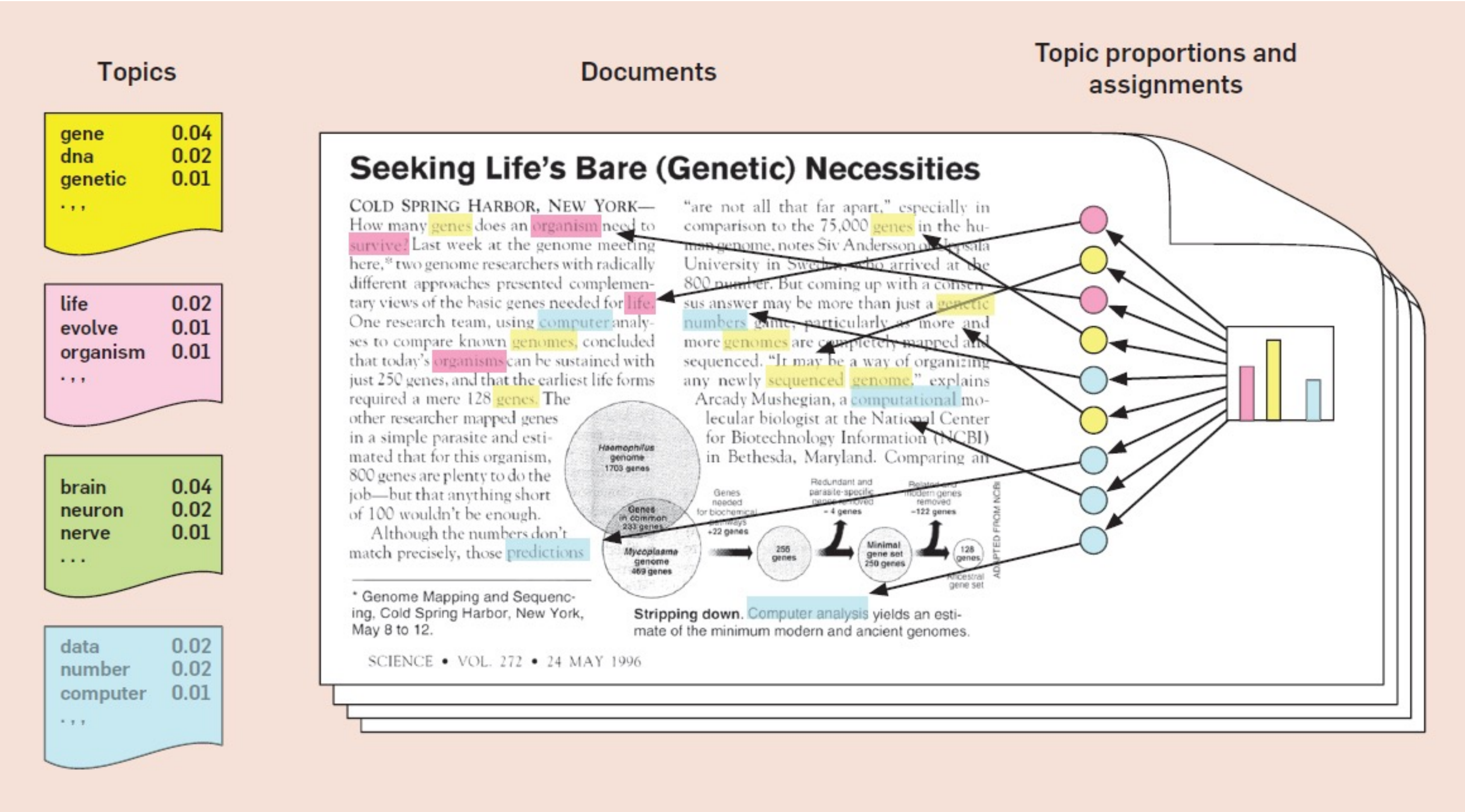
From: xor@clotho.acm.rpi.edu (Joe Schwartz)Subject: Re: NUTEK FACES APPLE'S WRATH (article!!!!!!) READ

In article <davea-120493231310@129.228.20.182> davea@xetron.com (David P. Alverson) writes:

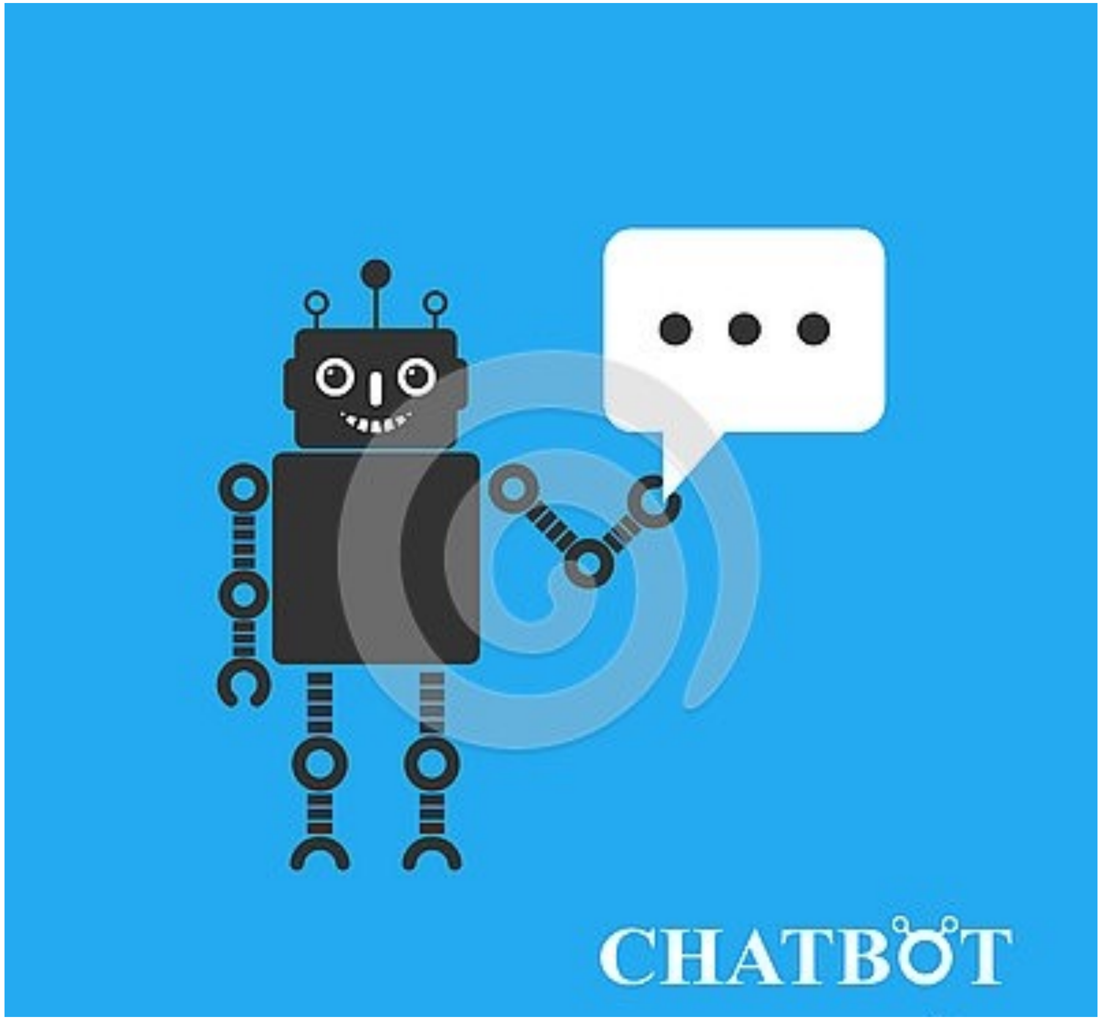
I believe Apple has a patent on the region features of QuickDraw. A mac clone would have to implement regions. This is why Apple's comment was that they believe it is not possible to make a Mac clone without infringing on their patents. They may have other patents like this.

Category: **Comp.mac.**

Applications



Applications



Applications

OpenAI

ChatGPT: Optimizing Language Models for Dialogue

We've trained a model called ChatGPT which interacts in a conversational way. The dialogue format makes it possible for ChatGPT to answer followup questions, admit its mistakes, challenge incorrect premises, and reject inappropriate requests. ChatGPT is a sibling model to InstructGPT, which is trained to follow an instruction in a prompt and provide a detailed response.

Applications

Write a Python script that does the following:

- loads a dataset from csv on disk to Pandas DataFrame

- ensures that dataset is entirely numeric; converts string values to numeric and tracks this conversion in a dictionary to reference later

- checks for missing values in the DataFrame and imputes missing values for any record that has one missing value; entirely drops a record that has more than one missing value

- splits the dataset into test and training sets

The Response

```
import pandas as pd

# Load dataset from csv to DataFrame
df = pd.read_csv('dataset.csv')

# Initialize dictionary to track string to numeric conversions
conversions = {}

# Convert string values to numeric and track conversions in dictionary
for col in df.columns:
    if df[col].dtype == object:
        conversions[col] = {val: i for i, val in enumerate(df[col].unique())}
        df[col] = df[col].map(conversions[col])

# Check for missing values
missing_values = df.isna().sum()
```

Applications

Nhờ chatGPT viết thư bày tỏ tình cảm với cô bạn gái cùng lớp

Yêu cầu: Viết cho tôi một lá thư để bày tỏ tình cảm của tôi với cô bạn gái cùng lớp. Sau đây là các đặc điểm của tôi.1: Tôi là một học sinh rất giỏi môn Tin học 2: Gia đình tôi là một gia đình truyền thống, bố mẹ tôi đều là giáo viên 3: Bố mẹ tôi thích bạn gái của tôi xuất thân từ một gia đình truyền thống, biết kính trên nhường dưới. 4: Tôi thích chơi đàn guitar cổ điển và có thể độc tấu guitar 5: Tôi yêu thích nhạc tình ca của Ngô Thụy Miên. 6: Tôi cũng yêu thích nhạc của Trịnh Công Son về thân phận con người và đât nước. 7:Tôi ít tiếp xúc với bạn gái 8: Bạn gái của tôi cũng học giỏi và rất nết na.

ChatGPT:

Kính gửi [Tên cô bạn gái],

Tôi viết thư này để bày tỏ cảm tình của tôi với cô. Tôi là một học sinh rất giỏi môn Tin học, và gia đình tôi là một gia đình truyền thống. Bố mẹ tôi đều là giáo viên và họ thích cô xuất thân từ một gia đình truyền thống, biết kính trên nhường dưới.

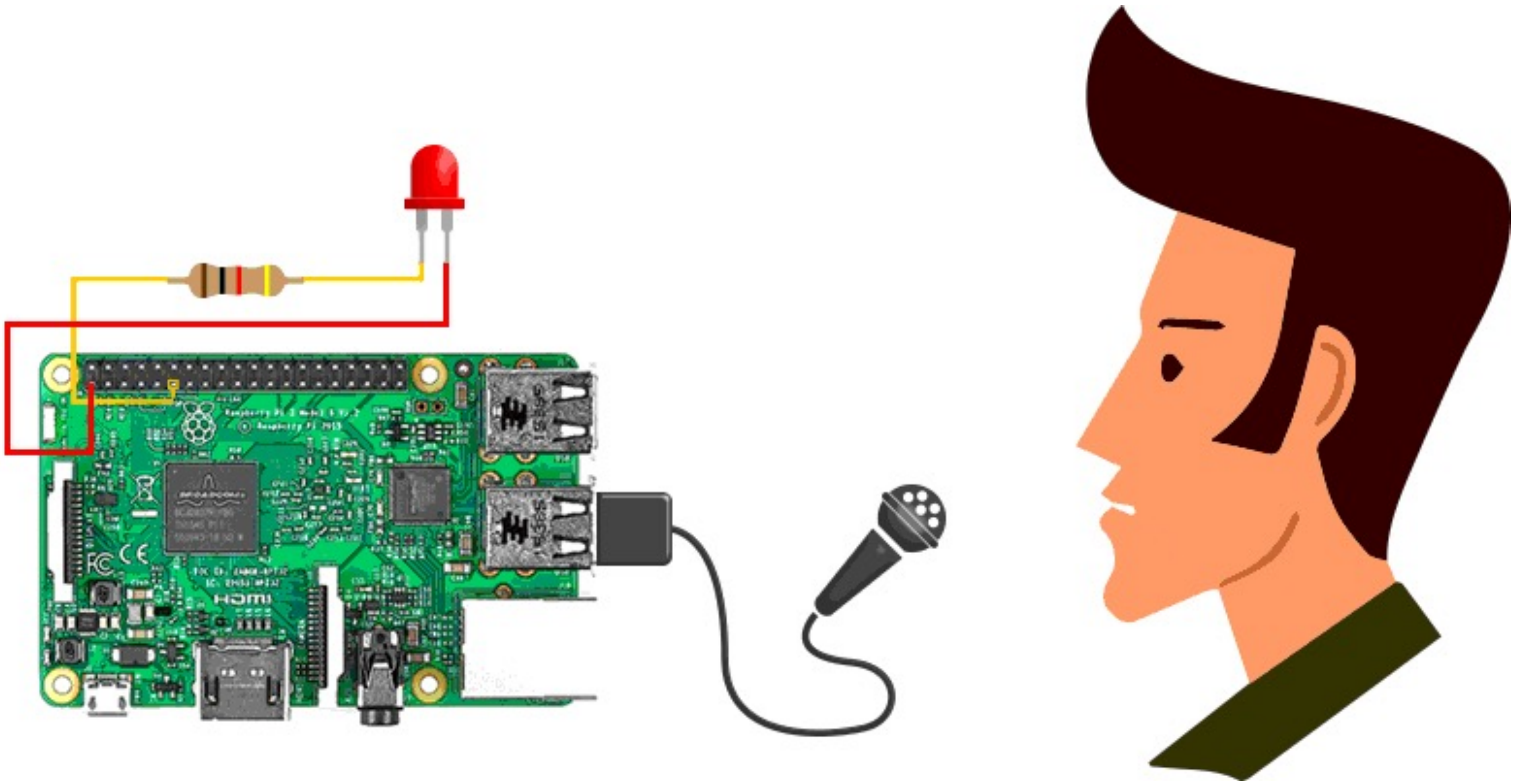
Ngoài việc học tập, tôi cũng thích chơi đàn guitar cổ điển và có thể độc tấu guitar. Tôi yêu thích nhạc tình ca của Ngô Thụy Miên và cũng rất thích nhạc của Trịnh Công Son về thân phận con người và đất nước.

Tôi biết rằng tôi chưa có nhiều cơ hội để tiếp xúc với cô, nhưng tôi muốn bắt đầu một mối quan hệ với cô. Tôi biết cô cũng là một học sinh rất giỏi và rất nết na, và tôi tin rằng chúng ta có thể học hỏi nhiều điều từ nhau.

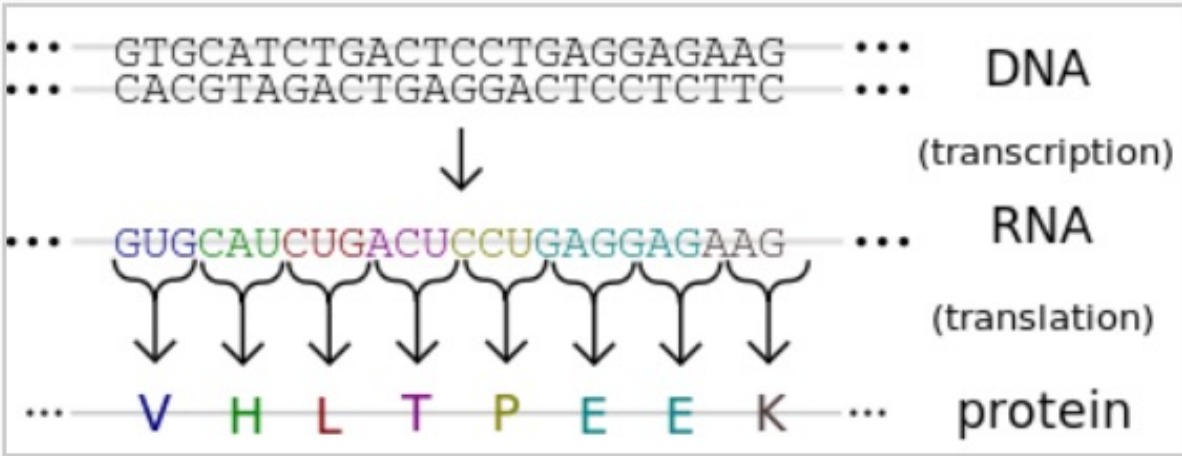
Tôi mong sẽ có cơ hội được gặp cô trong tương lai và tìm hiểu thêm về cô.

Trân trọng, [Tên của bạn]

Applications



Applications



Cancer?

Applications

Merci !